



main.R x cl_mma_codelist.csv x

```
1 CL_CODE_TYPE,CL_CODE,CL_MEDICATION_FULL,CL_MED_CLASS,CL_MED_ABBR
2 HCPCS,C9069,belantamab,ABCMA,BELA
3 HCPCS,J9037,belantamab,ABCMA,BELA
4 NDC,00173089601,belantamab,ABCMA,BELA
5 HCPCS,C9042,bendamustine,MUSTARD,BEND
6 HCPCS,C9243,bendamustine,MUSTARD,BEND
7 HCPCS,J9033,bendamustine,MUSTARD,BEND
8 HCPCS,J9034,bendamustine,MUSTARD,BEND
9 HCPCS,J9036,bendamustine,MUSTARD,BEND
10 NDC,10019007901,bendamustine,MUSTARD,BEND
11 NDC,16729025003,bendamustine,MUSTARD,BEND
12 NDC,16729025105,bendamustine,MUSTARD,BEND
13 NDC,42367052025,bendamustine,MUSTARD,BEND
14 NDC,42367052125,bendamustine,MUSTARD,BEND
15 NDC,60505622800,bendamustine,MUSTARD,BEND
16 NDC,63459034804,bendamustine,MUSTARD,BEND
17 NDC,63459039008,bendamustine,MUSTARD,BEND
18 NDC,63459039120,bendamustine,MUSTARD,BEND
19 NDC,63459039502,bendamustine,MUSTARD,BEND
20 NDC,63459039602,bendamustine,MUSTARD,BEND
21 NDC,68001057141,bendamustine,MUSTARD,BEND
22 NDC,68001057241,bendamustine,MUSTARD,BEND
23 NDC,71225012001,bendamustine,MUSTARD,BEND
24 HCPCS,C9207,bortezomib,PROTINHIB,BORT
25 HCPCS,J9041,bortezomib,PROTINHIB,BORT
26 HCPCS,J9044,bortezomib,PROTINHIB,BORT
27 HCPCS,J9046,bortezomib,PROTINHIB,BORT
28 HCPCS,J9048,bortezomib,PROTINHIB,BORT
29 HCPCS,J9049,bortezomib,PROTINHIB,BORT
30 HCPCS,50115,bortezomib,PROTINHIB,BORT
31 NDC,00143909801,bortezomib,PROTINHIB,BORT
32 NDC,00409170001,bortezomib,PROTINHIB,BORT
33 NDC,00409170301,bortezomib,PROTINHIB,BORT
530:1
```

219870_mm_optumlot : Dev

Stop

Sync All

Run job

File Edit Code View Plots Session Build Debug Profile Tools Help



main.R x cl_mma_rollup.csv x

Show whitespace

```

1 CL_MEDICATION_FULL,CL_MED_CLASS,CL_MED_ABBR,MONOMAIINTENANCE,DUALMAINTENANCEWITH,CONDITIONING,USED_FOR_OTHER_CANCERS
2 belantamab,ABCMa,BELA,,,,,
3 bendamustine,MUSTARD,BEND,,,,,YES - Mainly other cancers (Lymphoma). Used mono for MM
4 bortezomib,PROTINHIB,BORT,YES,LENA,,
5 carfilzomib,PROTINHIB,CARF,LENA,,
6 ciltacabtagene,ATCELL,CILT,,,,,
7 cisplatin,PLAT,CISP,,,,,YES - mainly others (solid)
8 cyclophosphamide,MUSTARD,CYCL,,,,,YES,YES - used in BC but also used in MM to similar frequency (maybe)
9 daratumumab,ACD38,DARA,YES,,
10 daratumumab_hyaluronidase,ACD38,DARA,YES,,
11 doxorubicin_peg_lip,TOPOINHIB,DOPL,,,,,YES - rarely MM, mainly solid"
12 doxorubicin,TOPOINHIB,DOXO,,,,,YES - rarely MM, mainly solid"
13 elotuzumab,ASLAF7,ELOT,,,,,
14 etoposide,TOPOINHIB,ETOP,,,,,YES - rarely MM, mainly solid"
15 idecabtagene,ATCELL,IDEC,,,,,
16 isatuximab,ACD38,ISAT,,,,,
17 ixazomib,PROTINHIB,IXAZ,YES,,
18 lenalidomide,IMMUNOMOD,LENA,YES,"BORT, CARF",YES - MDS (99% MM)
19 melphalan,MUSTARD,MELP,,,,,YES,
20 melphalan_betadex_sulf,MUSTARD,MELP,,,,,YES,
21 melphalan_diluent,MUSTARD,MELP,,,,,YES,
22 melphalan_fluf,MUSTARD,MELP,,,,,YES,
23 panobinostat,HIST,PANO,,,,,
24 pomalidomide,IMMUNOMOD,POMA,,,,,
25 selinexor,NUCLEAR,SELI,,,,,YES - mainly MM (some Lymphoma)
26 teclistamab,ABCMa,TECL,,,,,
27 thalidomide,IMMUNOMOD,THAL,,,,,YES - not really used at all in the US. Only approved for MM
28 elranatamab,ABCMa,ELRA,,,,,
29 linvoseltamab,ABCMa,LINV,,,,,
30 talquetamab,IMMUNOMOD,TALO,,,,,
31 venetoclax,BLC2I,VENE,,,,,YES - rarely use for MM as off-label use. Mainly other haem
32

```

32:1

Console Terminal Background Jobs

R - R 4.5.1 - /mnt/code/



main.R x cl_sct_codelist.csv x

```
1 CL CODE TYPE, CL_CODE, SCT_TYPE,
2 ICD10PROC, 30230G1, Alllogenic,
3 ICD10PROC, 30233G1, Alllogenic,
4 ICD10PROC, 30240G1, Alllogenic,
5 ICD10PROC, 30243G1, Alllogenic,
6 ICD10PROC, 30250G1, Alllogenic,
7 ICD10PROC, 30253G1, Alllogenic,
8 ICD10PROC, 30260G1, Alllogenic,
9 ICD10PROC, 30263G1, Alllogenic,
10 ICD10PROC, 30240G2, Alllogenic,
11 ICD10PROC, 30243G2, Alllogenic,
12 ICD10PROC, 30230G2, Alllogenic,
13 ICD10PROC, 30233G2, Alllogenic,
14 ICD10PROC, 30240G3, Alllogenic,
15 ICD10PROC, 30243G3, Alllogenic,
16 ICD10PROC, 30230G3, Alllogenic,
17 ICD10PROC, 30233G3, Alllogenic,
18 ICD10PROC, 30240G4, Alllogenic,
19 ICD10PROC, 30243G4, Alllogenic,
20 ICD10PROC, 30230G4, Alllogenic,
21 ICD10PROC, 30233G4, Alllogenic,
22 ICD9PROC, 41.02, Alllogenic,
23 ICD9PROC, 41.03, Alllogenic,
24 HCPCS, 38240, Alllogenic,
25 HCPCS, 52142, Alllogenic,
26 ICD10PROC, 30240X2, Alllogenic,
27 ICD10PROC, 30243X2, Alllogenic,
28 ICD10PROC, 30230X2, Alllogenic,
29 ICD10PROC, 30233X2, Alllogenic,
30 ICD10PROC, 30240Y2, Alllogenic,
31 ICD10PROC, 30243Y2, Alllogenic,
32 ICD10PROC, 30230Y2, Alllogenic,
33 ICD10PROC, 30233Y2, Alllogenic,
130:1
```

Console Terminal x Background jobs x

R 4.5.1 . /mnt/code/

[2026-04-14 15:30:22] =

[2026-04-14 15:30:22] Connected to Databricks



main.R x clintrial.csv x

```
1 code_type, code
2 HCPCS, G0276
3 HCPCS, G0292
4 HCPCS, G0293
5 HCPCS, G0294
6 HCPCS, G2000
7 HCPCS, G8928
8 HCPCS, G9057
9 HCPCS, G9927
10 HCPCS, S9988
11 HCPCS, S9990
12 HCPCS, S9991
13 HCPCS, S9992
14 HCPCS, S9994
15 HCPCS, S9996
16 ICD10DIAG, Z006
17 ICD9DIAG, V707
```



main.R x

diagnosthc_proc.csv x



1 proc_cd

2 C8900

3 C8901

4 C8902

5 C8903

6 C8904

7 C8905

8 C8906

9 C8907

10 C8908

11 C8909

12 C8910

13 C8911

14 C8912

15 C8913

16 C8914

17 C8918

18 C8919

19 C8920

20 C8921

21 C8922

22 C8923

23 C8924

24 C8925

25 C8926

26 C8927

27 C8928

28 C8929

29 C8930

30 C8931

31 C8932

32 C8933

33 C8934

327:1

219870_mm_optumlot : Dev

Stop

Sync All

Run job

- File
- Edit
- Code
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- Debug
- Profile
- Tools
- Help

main.R x mm_dx.csv x

```
1 icd_family, dx
2 ICD9DIAg, 203
3 ICD9DIAg, 203
4 ICD9DIAg, 20301
5 ICD9DIAg, 20302
6 ICD10DIAg, C900
7 ICD10DIAg, C9000
8 ICD10DIAg, C9001
9 ICD10DIAg, C9002
```



```
1 code_type,code
2 NDC,00003229111
3 NDC,00003452211
4 NDC,00015340420
5 NDC,00024065401
6 NDC,00024065601
7 NDC,00054038225
8 NDC,00054038325
9 NDC,00054412925
10 NDC,00054413025
11 NDC,00069303020
12 NDC,00069303120
13 NDC,00069303220
14 NDC,00069303320
15 NDC,00069303420
16 NDC,00069400405
17 NDC,00069401510
18 NDC,00069402625
19 NDC,00069403001
20 NDC,00069403101
21 NDC,00069403201
22 NDC,00069403301
23 NDC,00069403401
24 NDC,00069403701
25 NDC,00074056111
26 NDC,00074056114
27 NDC,00074056607
28 NDC,00074056611
29 NDC,00074057611
30 NDC,00074057622
31 NDC,00074057630
32 NDC,00074057634
33 NDC,00074057928
```

FileEditCodeViewPlotsSessionBuildDebugProfileToolsHelp



Go to file/functionAddins

main.R x mm_therapy.csv x

457	HCPCS, C9042
458	HCPCS, C9062
459	HCPCS, C9069
460	HCPCS, C9080
461	HCPCS, C9081
462	HCPCS, C9087
463	HCPCS, C9098
464	HCPCS, C9207
465	HCPCS, C9243
466	HCPCS, C9295
467	HCPCS, C9414
468	HCPCS, C9415
469	HCPCS, C9418
470	HCPCS, C9420
471	HCPCS, C9421
472	HCPCS, C9425
473	HCPCS, C9476
474	HCPCS, C9477
475	HCPCS, J8530
476	HCPCS, J8560
477	HCPCS, J8600
478	HCPCS, J9000
479	HCPCS, J9001
480	HCPCS, J9002
481	HCPCS, J9033
482	HCPCS, J9034
483	HCPCS, J9036
484	HCPCS, J9037
485	HCPCS, J9041
486	HCPCS, J9044
487	HCPCS, J9046
488	HCPCS, J9047
489	HCPCS, J9048



main.R x

other_malign.csv x

1 icd_family,dx,tumor_group

2 ICD9D1AG,141,Malignant neoplasm of base of tongue

3 ICD9D1AG,142,Malignant neoplasm of parotid gland

4 ICD9D1AG,143,Malignant neoplasm of upper gum

5 ICD9D1AG,144,Malignant neoplasm of anterior portion of floor of mouth

6 ICD9D1AG,145,Malignant neoplasm of cheek mucosa

7 ICD9D1AG,146,Malignant neoplasm of tonsil

8 ICD9D1AG,147,Malignant neoplasm of superior wall of nasopharynx

9 ICD9D1AG,148,Malignant neoplasm of postcricoid region of hypopharynx

10 ICD9D1AG,149,"Malignant neoplasm of pharynx, unspecified"

11 ICD9D1AG,150,Malignant neoplasm of cervical esophagus

12 ICD9D1AG,151,Malignant neoplasm of cardia

13 ICD9D1AG,152,Malignant neoplasm of duodenum

14 ICD9D1AG,153,Malignant neoplasm of hepatic flexure

15 ICD9D1AG,154,Malignant neoplasm of rectosigmoid junction

16 ICD9D1AG,155,"Malignant neoplasm of liver, primary"

17 ICD9D1AG,156,Malignant neoplasm of gallbladder

18 ICD9D1AG,157,Malignant neoplasm of head of pancreas

19 ICD9D1AG,158,Malignant neoplasm of retroperitoneum

20 ICD9D1AG,159,"Malignant neoplasm of intestinal tract, part unspecified"

21 ICD9D1AG,160,Malignant neoplasm of nasal cavities

22 ICD9D1AG,161,Malignant neoplasm of glottis

23 ICD9D1AG,162,Malignant neoplasm of trachea

24 ICD9D1AG,163,Malignant neoplasm of parietal pleura

25 ICD9D1AG,164,Malignant neoplasm of thymus

26 ICD9D1AG,165,"Malignant neoplasm of upper respiratory tract, part unspecified"

27 ICD9D1AG,170,"Malignant neoplasm of bones of skull and face, except mandible"

28 ICD9D1AG,171,"Malignant neoplasm of connective and other soft tissue of head, face, and neck"

29 ICD9D1AG,172,Malignant melanoma of skin of lip

30 ICD9D1AG,174,Malignant neoplasm of nipple and areola of female breast

31 ICD9D1AG,175,Malignant neoplasm of nipple and areola of male breast

32

1782.1

main.R x other_malign.csv x

```

1459 ICD10DIAG,C8319,"Mantle cell lymphoma, extranodal and solid organ sites"
1460 ICD10DIAG,C8330,"Diffuse large B-cell lymphoma, unspecified site"
1461 ICD10DIAG,C8331,"Diffuse large B-cell lymphoma, lymph nodes of head, face, and neck"
1462 ICD10DIAG,C8332,"Diffuse large B-cell lymphoma, intrathoracic lymph nodes"
1463 ICD10DIAG,C8333,"Diffuse large B-cell lymphoma, intra-abdominal lymph nodes"
1464 ICD10DIAG,C8334,"Diffuse large B-cell lymphoma, lymph nodes of axilla and upper limb"
1465 ICD10DIAG,C8335,"Diffuse large B-cell lymphoma, lymph nodes of inguinal region and lower limb"
1466 ICD10DIAG,C8336,"Diffuse large B-cell lymphoma, intrapelvic lymph nodes"
1467 ICD10DIAG,C8337,"Diffuse large B-cell lymphoma, spleen"
1468 ICD10DIAG,C8338,"Diffuse large B-cell lymphoma, lymph nodes of multiple sites"
1469 ICD10DIAG,C8339,"Diffuse large B-cell lymphoma, extranodal and solid organ sites"
1470 ICD10DIAG,C8350,"Lymphoblastic (diffuse) lymphoma, unspecified site"
1471 ICD10DIAG,C8351,"Lymphoblastic (diffuse) lymphoma, lymph nodes of head, face, and neck"
1472 ICD10DIAG,C8352,"Lymphoblastic (diffuse) lymphoma, intrathoracic lymph nodes"
1473 ICD10DIAG,C8353,"Lymphoblastic (diffuse) lymphoma, intra-abdominal lymph nodes"
1474 ICD10DIAG,C8354,"Lymphoblastic (diffuse) lymphoma, lymph nodes of axilla and upper limb"
1475 ICD10DIAG,C8355,"Lymphoblastic (diffuse) lymphoma, lymph nodes of inguinal region and lower limb"
1476 ICD10DIAG,C8356,"Lymphoblastic (diffuse) lymphoma, intrapelvic lymph nodes"
1477 ICD10DIAG,C8357,"Lymphoblastic (diffuse) lymphoma, spleen"
1478 ICD10DIAG,C8358,"Lymphoblastic (diffuse) lymphoma, lymph nodes of multiple sites"
1479 ICD10DIAG,C8359,"Lymphoblastic (diffuse) lymphoma, extranodal and solid organ sites"
1480 ICD10DIAG,C8370,"Burkitt lymphoma, unspecified site"
1481 ICD10DIAG,C8371,"Burkitt lymphoma, lymph nodes of head, face, and neck"
1482 ICD10DIAG,C8372,"Burkitt lymphoma, intrathoracic lymph nodes"
1483 ICD10DIAG,C8373,"Burkitt lymphoma, intra-abdominal lymph nodes"
1484 ICD10DIAG,C8374,"Burkitt lymphoma, lymph nodes of axilla and upper limb"
1485 ICD10DIAG,C8375,"Burkitt lymphoma, lymph nodes of inguinal region and lower limb"
1486 ICD10DIAG,C8376,"Burkitt lymphoma, intrapelvic lymph nodes"
1487 ICD10DIAG,C8377,"Burkitt lymphoma, spleen"
1488 ICD10DIAG,C8378,"Burkitt lymphoma, lymph nodes of multiple sites"
1489 ICD10DIAG,C8379,"Burkitt lymphoma, extranodal and solid organ sites"
1490 ICD10DIAG,C8380,"Other non-follicular lymphoma, unspecified site"
1491
1782:1
  
```

```
1 priginal_med,substitute_med
2 DARA,DARA
3 BORT,IXAZ
4 IXAZ,BORT
5
```



main.R x pregnancy.csv x



	code_type,code
1	ICD10PROC,102073Z
2	ICD10PROC,10207YZ
3	ICD10PROC,10900Z9
4	ICD10PROC,10900ZA
5	ICD10PROC,10900ZB
6	ICD10PROC,10900ZC
7	ICD10PROC,10900ZD
8	ICD10PROC,10900ZE
9	ICD10PROC,10900ZF
10	ICD10PROC,10900ZG
11	ICD10PROC,10900ZH
12	ICD10PROC,10900ZI
13	ICD10PROC,10900ZJ
14	ICD10PROC,10900ZK
15	ICD10PROC,10900ZL
16	ICD10PROC,10900ZM
17	ICD10PROC,10900ZN
18	ICD10PROC,10900ZO
19	ICD10PROC,10900ZP
20	ICD10PROC,10900ZQ
21	ICD10PROC,10900ZR
22	ICD10PROC,10900ZS
23	ICD10PROC,10900ZT
24	ICD10PROC,10900ZU
25	ICD10PROC,10900ZV
26	ICD10PROC,10900ZW
27	ICD10PROC,10900ZX
28	ICD10PROC,10900ZY
29	ICD10PROC,10900ZZ
30	ICD10PROC,10900ZA
31	ICD10PROC,10900ZB
32	ICD10PROC,10900ZC
33	ICD10PROC,10900ZD



424	ICD9DIAG, 6538
425	ICD9DIAG, 65381
426	ICD9DIAG, 65383
427	ICD9DIAG, 6539
428	ICD9DIAG, 65391
429	ICD9DIAG, 65393
430	ICD9DIAG, 6557
431	ICD9DIAG, 65571
432	ICD9DIAG, 65573
433	ICD9DIAG, 6558
434	ICD9DIAG, 65581
435	ICD9DIAG, 65583
436	ICD9DIAG, 6559
437	ICD9DIAG, 65591
438	ICD9DIAG, 65593
439	ICD9DIAG, 656
440	ICD9DIAG, 65601
441	ICD9DIAG, 65603
442	ICD9DIAG, 6563
443	ICD9DIAG, 65631
444	ICD9DIAG, 65633
445	ICD9DIAG, 6565
446	ICD9DIAG, 65651
447	ICD9DIAG, 65653
448	ICD9DIAG, 6566
449	ICD9DIAG, 65661
450	ICD9DIAG, 65663
451	ICD9DIAG, 6568
452	ICD9DIAG, 65681
453	ICD9DIAG, 65683
454	ICD9DIAG, 6569
455	ICD9DIAG, 65691
456	ICD9DIAG, 65693
5319.14	



main.R x pregnancy.csv x

687	HCPCS, 59840
688	HCPCS, 59841
689	HCPCS, 59850
690	HCPCS, 59851
691	HCPCS, 59852
692	HCPCS, 59855
693	HCPCS, 59856
694	HCPCS, 59857
695	HCPCS, 59866
696	HCPCS, 59870
697	HCPCS, 59871
698	HCPCS, 59897
699	HCPCS, 59898
700	HCPCS, 59899
701	HCPCS, 74712
702	HCPCS, 74713
703	HCPCS, 76801
704	HCPCS, 76802
705	HCPCS, 76805
706	HCPCS, 76810
707	HCPCS, 76811
708	HCPCS, 76812
709	HCPCS, 76813
710	HCPCS, 76814
711	HCPCS, 76815
712	HCPCS, 76816
713	HCPCS, 76817
714	HCPCS, 76818
715	HCPCS, 76819
716	HCPCS, 76820
717	HCPCS, 76821
718	HCPCS, 76825
719	HCPCS, 76826
5319:14	


```
846 ICD9DIA6,63502
847 ICD9DIA6,6351
848 ICD9DIA6,63511
849 ICD9DIA6,63512
850 ICD9DIA6,6352
851 ICD9DIA6,63521
852 ICD9DIA6,63522
853 ICD9DIA6,6353
854 ICD9DIA6,63531
855 ICD9DIA6,63532
856 ICD9DIA6,6354
857 ICD9DIA6,63541
858 ICD9DIA6,63542
859 ICD9DIA6,6355
860 ICD9DIA6,63551
861 ICD9DIA6,63552
862 ICD9DIA6,6356
863 ICD9DIA6,63561
864 ICD9DIA6,63562
865 ICD9DIA6,6357
866 ICD9DIA6,63571
867 ICD9DIA6,63572
868 ICD9DIA6,6358
869 ICD9DIA6,63581
870 ICD9DIA6,63582
871 ICD9DIA6,6359
872 ICD9DIA6,63591
873 ICD9DIA6,63592
874 ICD9DIA6,636
875 ICD9DIA6,63601
876 ICD9DIA6,63602
877 ICD9DIA6,6361
```

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2036 ICD10D1AG,0083
2037 ICD10D1AG,0084
2038 ICD10D1AG,0085
2039 ICD10D1AG,0086
2040 ICD10D1AG,0087
2041 ICD10D1AG,00881
2042 ICD10D1AG,00882
2043 ICD10D1AG,00883
2044 ICD10D1AG,00889
2045 ICD10D1AG,0089
2046 ICD10D1AG,00900
2047 ICD10D1AG,00901
2048 ICD10D1AG,00902
2049 ICD10D1AG,00903
2050 ICD10D1AG,00910
2051 ICD10D1AG,00911
2052 ICD10D1AG,00912
2053 ICD10D1AG,00913
2054 ICD10D1AG,009211
2055 ICD10D1AG,009212
2056 ICD10D1AG,009213
2057 ICD10D1AG,009219
2058 ICD10D1AG,009291
2059 ICD10D1AG,009292
2060 ICD10D1AG,009293
2061 ICD10D1AG,009299
2062 ICD10D1AG,00930
2063 ICD10D1AG,00931
2064 ICD10D1AG,00932
2065 ICD10D1AG,00933
2066 ICD10D1AG,00940
2067 ICD10D1AG,00941
5319:14



R - R 4.5.1 . /mnt/code/

Console

Terminal

Background Jobs

[2026-04-14 15:30:22]

=====

[2026-04-14 15:30:22] Connected to Databricks



5287 ICD9DIAG,Z0542
5288 ICD9DIAG,Z0542
5289 ICD9DIAG,Z0543
5290 ICD9DIAG,Z055
5291 ICD9DIAG,Z056
5292 ICD9DIAG,Z0571
5293 ICD9DIAG,Z0572
5294 ICD9DIAG,Z0573
5295 ICD9DIAG,Z058
5296 ICD9DIAG,Z059
5297 ICD9DIAG,Z364
5298 ICD9DIAG,Z3681
5299 ICD9DIAG,Z3683
5300 ICD9DIAG,Z3684
5301 ICD9DIAG,Z3688
5302 ICD9DIAG,Z3800
5303 ICD9DIAG,Z3801
5304 ICD9DIAG,Z381
5305 ICD9DIAG,Z382
5306 ICD9DIAG,Z3830
5307 ICD9DIAG,Z3831
5308 ICD9DIAG,Z384
5309 ICD9DIAG,Z385
5310 ICD9DIAG,Z3861
5311 ICD9DIAG,Z3862
5312 ICD9DIAG,Z3863
5313 ICD9DIAG,Z3864
5314 ICD9DIAG,Z3865
5315 ICD9DIAG,Z3866
5316 ICD9DIAG,Z3868
5317 ICD9DIAG,Z3869
5318 ICD9DIAG,Z387
5319 ICD9DIAG,Z388
5319:14